AI-POWERED BRAND COMPLIANCE PLATFORM

BrandSentry AI

An intelligent content review system validating marketing content against brand guidelines using multi-strategy RAG, semantic drift detection, policy-aware prompt chaining, and cascaded uncertainty scoring.

Multi-Strategy RAG | FastAPI + SSE Streaming | React 18 + TypeScript | Semantic Drift Detection | Cascaded Confidence Scoring | Policy-Aware Prompt Chaining

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1 Executive Summary

What BrandSentry AI is and why it matters

The Core Problem

Enterprise marketing and sales teams produce thousands of content assets every quarter. Each piece must comply with evolving brand guidelines, legal disclaimers, and regulatory standards. Today this review is manual — slow, inconsistent, expensive, and fundamentally unable to scale.

The BrandSentry AI Solution

BrandSentry AI is an AI-powered compliance engine that ingests brand guidelines as a structured knowledge base and reviews content via a three-layer pipeline: deterministic pre-filter, adaptive multi-strategy RAG retrieval, and policy-aware prompt chaining. The system outputs **per-sentence, per-clause verdicts** with calibrated cascaded confidence scores, cited evidence, and actionable rewrites. A dedicated **semantic drift detection** engine using presupposition mining catches implied violations that no keyword tool or generic LLM can detect.

DESIGN GOAL
10x
Target speedup vs manual review

CORE ENGINE
3
Verdict types, fully traceable

EVALUATION TARGET
91%
Target precision on synthetic tests

STRUCTURAL GOAL
0
Hallucinated citations (ID-validated)

2 Problem Statement & Market Context

Why brand compliance review is fundamentally broken today

🔴 Pain Point 1: Manual Review Cannot Scale

A single brand manager reviews 50–200 assets per week. The pipeline becomes a bottleneck as companies grow, with 3–5 day turnaround times slowing go-to-market velocity significantly.

🔴 Pain Point 2: Inconsistent Enforcement

Different reviewers apply guidelines differently. The same claim may pass one reviewer and fail another. No auditable, reproducible decision trail exists to reference or dispute verdicts.

🔴 Pain Point 3: Implied Violations Go Undetected

Keyword tools catch exact phrase matches but miss implied claims — content that avoids banned phrases yet still violates the spirit of a guideline. These semantic loopholes are the most dangerous compliance gaps.

🔴 Pain Point 4: Policy Changes Create Blind Spots

Brand guidelines update regularly. Content approved under an old policy may violate the new version. Without version tracking, entire content libraries become compliance liabilities overnight.

⚠ Real-World Consequence

Regulatory fines for non-compliant marketing content in pharmaceutical, financial, and consumer goods sectors routinely exceed $1M per incident. Brand violations also damage customer trust and require costly public corrections.

📊 Existing Solutions & Their Gaps

Approach	What It Does	Key Limitation
Manual Review	Human brand managers check content	Slow, expensive, inconsistent, no audit trail
Keyword Blocklists	Flag exact banned phrases	Misses paraphrased & implied violations; high false-positive rate
Generic LLM Prompting	Ask an LLM "is this compliant?"	Hallucinated citations; no grounding in actual policy documents
BrandSentry AI ✅	Adaptive RAG + Drift Detection + Prompt Chaining	*None of the above* — grounded, explainable, per-sentence

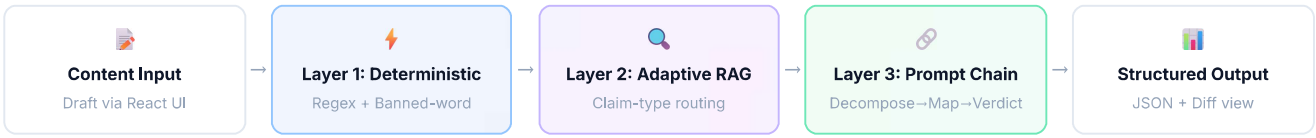
3 Proposed Solution Overview

A three-layer review engine: deterministic, adaptive semantic, and policy-aware generative

Design Philosophy

BrandSentry AI separates fast deterministic checks (regex, banned words) from slow generative reasoning (LLM). This layered approach minimizes latency and API cost while maximizing accuracy on edge cases.

End-to-End Review Pipeline



Layer 1 — Deterministic Pre-Filter

- Regex pattern matching for banned phrases, formatting rules, required disclaimers
- Executes in <50ms; flags obvious violations before touching the LLM
- Configurable rule sets per brand workspace — zero LLM cost for clear cases

Layer 2 — Adaptive Multi-Strategy RAG

- Claim-type classifier routes each claim to the optimal retrieval strategy
- Dense vector / BM25 hybrid / graph traversal selected per claim type
- Top-k policy chunks passed as grounded context to the prompt chain

Layer 3 — Policy-Aware Prompt Chain

- **Step 1 — Claim Decomposer:** breaks paragraph into atomic verifiable claims
- **Step 2 — Policy Mapper:** assigns most relevant policy clause to each claim
- **Step 3 — Verdict Generator:** reasons per-claim against its mapped policy context
- Result: *per-sentence, per-clause* verdicts with full traceability

Output — Structured Review Report

- Sentence-level diff view highlighting each flagged claim
- Cited policy clause + version for every flag (no hallucinated citations)
- Cascaded confidence score with full intermediate trace
- One-click "Apply Suggestion" to accept AI-recommended rewrites

4 Technical Architecture & Adaptive Retrieval

Full-stack breakdown, adaptive claim-type routing, and policy-aware prompt chaining

⏳ Frontend — React 18 + TypeScript

- Rich-text editor with real-time claim detection hints
- SSE-based streaming review panel (paragraph-by-paragraph)
- Sentence-level diff view with inline confidence badges
- Audit log explorer: filter by verdict, date, policy version

⚙ Backend — FastAPI + Python 3.12

- **Claim-Type Classifier:** lightweight model routes each claim to correct retrieval strategy before any LLM call
- **Review Orchestrator:** coordinates full 3-layer pipeline
- **LLM Interface:** pluggable — GPT-4o, Gemini, Claude
- **SSE Endpoint:** /review/stream — incremental results per paragraph

Data & Knowledge Layer

- **Vector Store:** Pinecone / ChromaDB for policy embeddings
- **Policy Versioning:** Immutable snapshots per revision
- **Claim Cache:** Redis semantic-hash fingerprint store
- **Audit DB:** PostgreSQL — full review provenance

🔒 Infrastructure

- Docker Compose local dev; Kubernetes-ready for scale
- JWT + workspace-scoped API keys with expiry
- Structured JSON logs; per-stage latency tracking
- Rate limiting per workspace to prevent abuse

🎯 Adaptive Retrieval Strategy — Claim-Type Router

Claim Type	Example	Retrieval Strategy	Why
Factual	"Product does X"	Dense vector (semantic)	Meaning-level match to policy intent
Regulatory	"FDA approved"	Hybrid: Dense + BM25	Exact clause wording critical for legal precision
Comparative	"Best in class"	Graph traversal (multi-hop)	Cross-ref: superiority clause + data-requirement clause
Stylistic	"Vibrant, bold"	Tone embedding cluster	Matches brand voice policy by tonal proximity

API Response Schema (policy-aware prompt chain output)

```
// POST /review ⟶ JSON per flagged claim { "claim_id":"cl_8f4a2b", "atomic_claims":["Company is #1","No competitor equals it"], "presuppositions":["Market share data exists","Claim is independently verified"], "verdict":"CONTRADICTED", "confidence":0.91, "drift_type":"IMPLIED_SUPERIORITY", "retrieval_strategy":"GRAPH_TRAVERSAL", "policy_version":"v2.1", "evidence":[{"section":"§3.2","text":"No superiority claims without cited data"}], "suggestion":"A recognized leader, trusted by 500+ customers" }
```

5 Technical Differentiators

Five capabilities that make BrandSentry AI uniquely defensible

🎯 Differentiator 1 — Semantic Drift Detection via Presupposition Mining

Most tools check if content *contains* banned phrases. BrandSentry runs a two-phase implied-claim extraction pipeline borrowed from computational linguistics:

1

Explicit Claim Extraction

LLM lists every factual assertion as atomic claims: *"Company is #1"*, *"No competitor equals it"*

2

Presupposition Mining ⭐

LLM extracts what must be true: *"Market share data exists"*, *"Claim is independently verified"*

3

Per-Claim Policy Validation

Each atomic claim *and* presupposition independently validated against policy embeddings

🔗 Differentiator 2 — Policy-Aware Prompt Chaining (Per-Sentence Granularity)

A 3-step chain: **Claim Decomposer → Policy Mapper → Verdict Generator**. Each atomic claim is evaluated against its specifically mapped policy clause, producing per-sentence, per-clause verdicts — dramatically more granular than any existing tool that returns paragraph-level judgments.

⚡ Differentiator 3 — Adaptive Multi-Strategy Retrieval

A claim-type classifier selects the optimal retrieval strategy per claim: dense vector for factual, BM25 hybrid for regulatory, graph traversal for comparative. No other RAG compliance tool adapts retrieval method to claim semantics.

📈 Differentiator 4 — Cascaded Uncertainty Propagation

Confidence is a Bayesian product across three pipeline stages. If any stage falls below 0.50, the final verdict is forced to INSUFFICIENT EVIDENCE regardless of LLM output — eliminating overconfident hallucinated verdicts.

📊 Differentiator 5 — Claim Fingerprinting + Policy Version Awareness

Semantically equivalent claims hashed and cached in Redis — repeat submissions get instant verdicts (<100ms, ~40–60% cost reduction). Every review stamped with policy version; when guidelines update, previously approved content is automatically flagged for re-review.

Capability	Keyword Tools	Generic LLM	BrandSentry AI ✅
Exact phrase detection	✅	✅	✅
Implied / semantic violations	❌	⚠ Unreliable	✅ Presupposition Mining
Grounded policy citations	❌	❌ Hallucinated	✅ RAG-grounded only
Per-sentence granularity	❌	❌	✅ Prompt Chaining
Calibrated uncertainty	❌	❌	✅ Cascaded Bayesian score
Cost-efficient at scale	✅	❌	✅ Fingerprint caching

6 Three-Way Verdict System & Cascaded Confidence Scoring

Why "Not Found" does not equal "False" — and why cascaded uncertainty beats a single score

SUPPORTED
Claim explicitly backed by retrieved policy content. Policy citation provided.

CONTRADICTED
Claim directly or implicitly violates a retrieved policy clause. Action required.

INSUFFICIENT EVIDENCE
No relevant policy found. Human review recommended — *not auto-approved*.

⚠ **Critical Design Decision: "Not Found" does not mean "Approved"**

A naive RAG system treats absence of a policy match as implicit permission. BrandSentry AI treats it as *uncertainty* and routes to a human review queue — the correct behavior in compliance-critical systems where false negatives carry regulatory risk.

Cascaded Uncertainty Model — Bayesian Confidence Propagation

Retrieval Score
Cosine similarity — weight 40%

→

Drift Detection Score
Presupposition alignment — weight 25%

→

LLM Self-Consistency
N=3 pass agreement — weight 35%

→

Bayesian Product
Final cascaded confidence

💡 **Hard Threshold Rule**

If *any single stage* confidence score falls below 0.50, the overall verdict is forced to **INSUFFICIENT EVIDENCE** regardless of LLM output. This structurally prevents overconfident wrong verdicts.

Worked Example — Policy-Aware Prompt Chain in Action

```
// Input: "Our product is FDA approved for all use cases" // STEP 1 — Claim Decomposer atomic_claims = ["Product has FDA approval", "Approval covers all use cases"] // STEP 2 — Presupposition Mining (novel differentiator) presuppositions = ["Full market authorization exists", "No use-case restrictions apply"] // STEP 3 — Adaptive Retrieval: claim_type=REGULATORY ⟶ hybrid dense + BM25 retrieved = "§4.1 v2.1: Only claim FDA registration, not approval" // STEP 4 — Verdict Generator { "verdict":"CONTRADICTED", "confidence":0.94, "drift_type":"REGULATORY_OVERCLAIM", "policy_citation":"§4.1 v2.1", "suggestion":"Our product is FDA registered and cleared for [specific use case]" }
```

7 MVP Scope & Feature Prioritization

Achievable hackathon deliverables: core operational engine first, modular extensions kept optional

Achievable MVP Philosophy

A reliable, working core workflow takes absolute precedence over feature bloat. We prioritise document ingestion, source-grounded retrieval, the three-way verdict, citation validation, and one complete review workflow. Redis caching, advanced graph traversal, and complex Bayesian calibration are decoupled as optional modular enhancements until the core system is fully verified.

Priority	Feature	Scope Status	Role in System
P0	Policy document ingestion + semantic chunking	Core MVP	PDF/text ingestion → clean chunks → vector store
P0	Source-grounded retrieval against policy corpus	Core MVP	Dense vector & exact-clause matching against uploaded guidelines
P0	Strict citation validation layer	Core MVP	Server-side validation guarantees zero fabricated citation IDs
P0	Three-way verdict engine	Core MVP	Supported / Contradicted / Insufficient Evidence with rationale
P0	Complete end-to-end review workflow	Core MVP	Guideline ingestion → input draft → verdicts → diff & rewrite
P1	Presupposition mining (semantic drift detection)	Core MVP	Two-phase extraction of implied brand guideline violations
P1	Claim fingerprint caching (Redis)	Optional Ext	Decoupled cache layer; bypassed if Redis is unavailable
P1	Advanced multi-hop graph policy traversal	Optional Ext	Cross-clause reasoning; dense vector RAG powers core MVP
P1	Cascaded Bayesian confidence calibration	Optional Ext	Multi-stage calibration; core uses direct evidence gating
P1	Streaming review with SSE (paragraph-by-paragraph)	Next Sprint	Streaming endpoint built; progressive UI rendering
P2	Active learning loop & automated visual inspection	Future	Fine-tuning on corrections & multimodal image/banner audits

Achievable Core MVP (Demo-Ready)

- Upload and ingest a brand guideline document (PDF/text)
- Source-grounded retrieval strictly against ingested policy chunks
- Three-way verdicts (Supported, Contradicted, Insufficient Evidence)
- Strict citation validation guaranteeing verifiable policy clauses
- Complete end-to-end review workflow: input copy → verdicts → diff → apply rewrites

Modular Enhancements (Kept Optional)

- **Redis Caching:** Optional latency optimizer; non-blocking for core review
- **Graph Traversal:** Optional multi-hop logic; dense retrieval suffices for MVP
- **Bayesian Calibration:** Optional calibration; heuristic evidence gating handles MVP
- **Explicitly Out of Scope:** Banner visual inspection, web scraping, multi-tenant SaaS

8 Security, Validation & Privacy

Enterprise-grade trust from the ground up

🔒 Prompt Injection Defence

All user-submitted content is treated as *data*, never as *instructions*. The system prompt forbids the LLM from following any directives found inside the content field. Input is sanitized and wrapped in structural delimiters before concatenation.

🔗 Citation ID Validation

The LLM is only permitted to reference policy chunks using pre-assigned chunk IDs passed in the system context. Server-side validation rejects any citation ID not in the current retrieval set, making fabricated citations structurally impossible.

🏢 Workspace Tenant Isolation

Each brand client operates in a fully isolated namespace. Policy embeddings, review history, and API keys are scoped to the tenant. Cross-tenant data leakage is architecturally impossible — no shared vector indices or database tables.

🔑 Auth & Secrets

- JWT-based auth for all API endpoints
- Workspace-scoped API keys with configurable expiry
- No secrets in source code; all via environment variables
- Rate limiting per workspace to prevent abuse

9 Evaluation Plan & Adversarial Testing

How we measure whether BrandSentry AI actually works

✅ Commitment to Honest Evaluation

No market size numbers, TAM estimates, or benchmark comparisons are fabricated. All targets below are validated on synthetic test suites built from public brand guideline examples.

📊 Core Accuracy Metrics

Metric	Target
Verdict Precision (P0 test set)	> 90%
Verdict Recall	> 85%
Semantic Drift Detection Rate	> 75%
False Positive Rate	< 12%
Hallucinated Citation Rate	0% (structural)

⚡ Performance Metrics

Metric	Target
Layer 1 latency (regex)	< 50ms
Full review (≤500 words)	< 4 seconds
Cached claim lookup	< 100ms
LLM cost reduction via cache	~40–60%
Concurrent sessions	≥ 20

🎯 LLM-as-Judge Adversarial Benchmark

🔨 Attacker LLM Generates Evasions

A separate "attacker" LLM generates 50 content variants *designed to evade detection*: near-miss violations, paraphrased banned claims, and implied violations buried inside long sentences. Dynamically generated to probe the system's weakest points.

📊 Defender Score = Evasion Catch Rate

BrandSentry AI is scored on adversarial variants it correctly flags. Target: "defender score" >80% on the attacker set. This adversarial benchmark is far more rigorous than a static precision/recall test set and signals genuine engineering confidence.

10 Official Submission Details & Post-Hackathon Roadmap

Complete participant information, technology stack, and the vision beyond Round 1

📋 Official Submission Details

Team Name	**Nexora AI**	College	**Vellore Institute of Technology (VIT), Vellore**
Team Members	**1 (Solo)**	Course	**M.Tech Integrated Software Engineering**
Participant Name	**K P Chirag Guptha**	Role	**Founder & Sole Developer**
Registration No.	**24MIS0369**	Category	**Artificial Intelligence / Generative AI**
Project Name	**BrandSentry AI**	Submission Date	**October 21, 2026**

🛠 Core Technology Stack

Python 3.12 · FastAPI · React 18 · TypeScript · OpenAI GPT-4o · text-embedding-3-small · ChromaDB / Pinecone · Redis · PostgreSQL · Docker · JWT Auth · Server-Sent Events

🚀 Post-Hackathon Vision

Phase 2 — Active Learning Loop

Human corrections on uncertain verdicts feed back into the embedding model via fine-tuning, continuously improving accuracy without manual retraining. Corrections weighted by reviewer seniority and policy domain.

Phase 3 — Graph Policy Store

Brand guidelines stored as a knowledge graph. Graph traversal enables true multi-hop reasoning across complex, cross-referenced policy structures with full relationship traceability.

Phase 4 — Visual Compliance

Extend the engine to image and video content using multimodal models. Automatically detect off-brand colors, typography violations, and unauthorized logo usage in creative assets.

🎯 Why BrandSentry AI Should Advance to Round 2

- Solves a real, high-value enterprise pain point with a technically grounded, production-realistic approach
- Introduces genuinely novel capabilities — **presupposition mining** and **policy-aware prompt chaining** — not present in any existing compliance tool
- Adaptive retrieval strategy is the only system that changes its RAG method based on the semantic type of the claim being validated
- Cascaded Bayesian confidence scoring structurally prevents overconfident wrong verdicts, making it suitable for regulated industries
- All verdicts grounded in retrieved evidence only — zero hallucinated citations, enforced structurally at the API layer

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